

Trunk Emblem Display

A round 2.8" ESP32 touchscreen in a 3D-printed 88 mm housing that sits over the trunk badge holder. It has an open back so it stands just 10.3 mm off the paint, locates on the factory peg holes, takes its wires through one 16 mm hole drilled at the badge centre, runs from an always-on 12 V fuse with battery protection, and shows the classic roundel or your own designs, controlled from your phone.

Everything is at github.com/cakemasterjedi/haloweb

Install the firmware: cakemasterjedi.github.io/haloweb. Print files are in the **hardware** folder, and this guide and the wiring diagram are in **docs**. Green **Code** button → **Download ZIP** gets it all at once.



88 mm

outside diameter (fits over the 84 mm badge holder)

10.3 mm

off the paint, including the tape

16 mm

one hole drilled at the badge centre for the wires

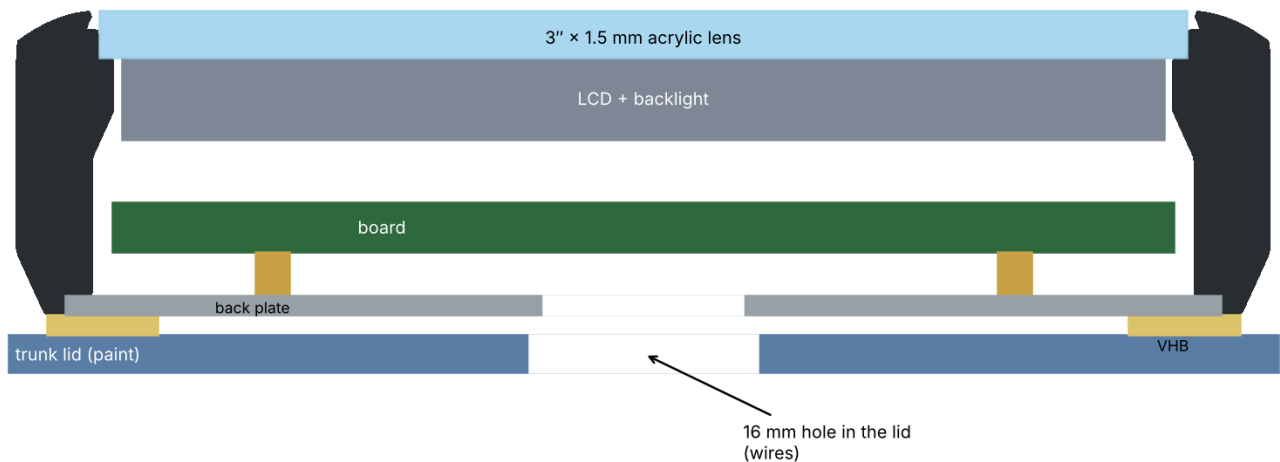
Contents

- 1 Before you start
- 2 Parts and tools
- 3 Printing
- 4 Firmware
- 5 Wiring overview
- 6 Build the display unit
- 7 Build the trunk-lid tray
- 8 Install on the car
- 9 Test and calibrate
- 10 Troubleshooting
- A Heat spreader and thermal pads

1 Before you start

The build has two units. The **display unit** replaces the badge: a printed 88 mm housing holding the Waveshare board, with a clear 3" acrylic lens flush with the rim. The display goes in **from the back**: the board drops into a wide pocket and the glass stops against a lip at the front, and the top of the housing covers the screen's cable tab, so from the front you see a clean circle. A thin printed **back plate** closes the back. It drops into the base, rests on the board's brass studs and sits flush with the base, and its two short solid pegs locate in the factory badge holes. The wires leave through a hole in the middle of the plate and one **16 mm hole drilled at the badge centre**. The **trunk-lid tray** sits behind the lid's inner trim and holds the 12 V → 5 V converter and the INA219 voltage sensor.

Section through the screen centre (heights × 2.2)



How the parts stack (heights exaggerated). The glass sits against the front lip, the board's brass studs rest on the 0.6 mm back plate, the housing base and plate edge sit on a 0.64 mm VHB ring inside the badge holder, and the 1.5 mm lens sits on the glass, flush with the rim. The screen sits 2.65 mm above the housing's centre under a centred lens; that is what lets its cable tab fit inside 88 mm.

Read these first

- **Battery:** disconnect the 12 V battery (in the trunk on the G30) before any wiring, and tap an always-on circuit only with the firmware's battery protection turned on (section 4). Fit the kill switch if the car sits for weeks.
- **Legal:** rules on lit or animated displays facing backward vary by state and country. Check yours; a dim or static image is the safe choice while driving.
- **Sealing:** the display is sealed shut with silicone. Install the firmware and check the phone control page works before step 6.10 (bonding the lens), because the USB port can't be reached afterwards.
- **Drilling:** this version needs one 16 mm hole in the trunk lid at the badge centre (section 8). Check behind the panel before you drill.
- **Heat:** print the final parts in ASA. PLA and PETG soften on a sunny trunk lid. For extra cooling, see Appendix A.

Unofficial fan project, not affiliated with or endorsed by BMW. Work at your own risk.

2 Parts and tools

PART	QTY	NOTES
☐ Waveshare ESP32-S3-Touch-LCD-2.8C	1	73 mm round glass, 480×480. Comes with 12-pin and 4-pin SH1.0 cables.
☐ 3" × 1/16" clear acrylic disc	1	Enoin 3" × 0.060" (76.2 × 1.5 mm). The lens; it sits flush in a 1.5 mm pocket.
☐ INA219 module (GY-219)	1	25 × 22 mm. Reads the car's voltage over I2C.
☐ YRDZXG 12/24 V → 5 V 5 A 25 W buck converter	1	Sealed. Body about 46 × 32 × 18 mm, 61 mm over the mounting ears. Red/black in, yellow/black out.
☐ 28 AWG silicone wire, red + black	~1 m each	5 V and ground. All six display wires leave through the centre hole.
☐ 30 AWG silicone wire, 3 colors	~1 m each	3.3 V, GND, SDA, SCL (4 wires).
☐ 20 AWG automotive wire, red + black	~3 m each	12 V feed from the fuse box to the trunk lid.
☐ Fuse tap kit (Mini and ATO) + 3 A fuses	1	The G30 fuse box has both Mini and ATO (standard-size) blade fuses. Get a kit with both tap sizes and 3 A fuses in both sizes; you use the one that matches the fuse you tap.
☐ 12 V kill switch, 5 A or more (optional)	1	Inline rocker on a pigtail, or a small sealed toggle. Turns the display fully off for storage.
☐ 16 mm (5/8") step drill + 5/8" rubber grommet	1	The wire hole at the badge centre. Plus touch-up paint for the bare edge.
☐ Ring terminal (M6 / ¼")	1	Chassis ground.
☐ Solder-seal heat-shrink connectors	~10	Waterproof wire joins.
☐ LOCA UV-cure optical glue	1	Bonds the lens to the screen glass. Not B-7000 or super glue: they craze acrylic.
☐ Permatex Ultra Black (82180)	1	Sensor-safe, neutral-cure silicone for all seals.
☐ 3M VHB 5925 (0.64 mm), black, 1"	1 roll	Cut into a ring with the template in section 8. It sits inside the 84 mm badge holder under the housing base and plate edge.
☐ 3M adhesion promoter, isopropyl alcohol	—	Surface prep for the VHB.
☐ Mini silica gel pack	1	Taped to the back of the board.
☐ Conformal coating, e.g. MG Chemicals 422C (silicone) or 419D (acrylic)	1 small can or pen	Coats the boards so a drop of water or condensation can't short them. See step 6.4.
☐ Small UV-resistant zip ties (≤4.8 mm)	~8	Tray straps and harness ties.
☐ Tesa cloth tape, split loom 6–8 mm	—	Make the new wiring look factory and stop rattles.
☐ Optional heat spreader: 0.016" (0.4 mm) aluminium sheet, soft thermal pads 0.5–1.5 mm, Kapton tape	—	See Appendix A.

☐ ASA filament (black) + PLA for test fit — Polymaker, Bambu or eSun ASA.

Tools: 3D printer (enclosed for ASA), soldering iron, wire strippers for 20–30 AWG, heat gun, multimeter, trim removal tools, painter's tape, UV flashlight (for LOCA), masking tape, USB-C cable and a computer with Arduino IDE.

3 Printing

FILE	ORIENTATION	PURPOSE
HaLo_88_HOUSING.3mf	Back down (as supplied)	The housing. The display goes in from the back.
HaLo_88_BACK_PLATE.3mf	Flat, pegs up (as supplied)	0.6 mm plate that closes the back: centre hole for the wires, two 6 mm locating pegs, shallow dome under the paint side
HaLo_88_BACK_PLATE_SPREADER.3mf	Flat, pegs up	Instead of the plain plate, if you fit the aluminium heat spreader (Appendix A)
trunk_electronics_tray.3mf	Flat, back down	Holds the converter and INA219 in the trunk lid

Settings (ASA)

- 0.2 mm layers, **5 walls, 6 top / 5 bottom layers**, 40 % gyroid infill, flow about 2–5 % above your normal setting. Thick, slightly over-filled walls are what make an FDM print watertight; with the usual 2–3 walls, water can seep through the layer lines.
- **Clean walls:** outer wall about 40 mm/s, inner walls first, part cooling off for the first 3 layers and low (10–20 %) after that. If a few layers look frayed or stringy, dry the filament (ASA picks up moisture) and slow the outer wall down; the housing's thin wall at 6 o'clock (behind the cable ribbon, 0.8 mm) is the part that shows it first.
- **No brim or supports needed.** The housing's bottom edge is a 45° bevel and the board-pocket roof is a 45° slope, so nothing overhangs. Use a clean textured PEI or glue-stick bed at 100–110 °C with the enclosure closed.
- Back plate: 0.2 mm layers, 3 walls, 100 % infill so the pegs are solid. Let it cool on the bed so it stays flat.

Test fit in PLA first

- 1 **Make a jig:** drill two 5 mm holes 56 mm apart and a 16 mm hole midway between them through a flat scrap board. It stands in for the trunk lid during the test fit and assembly.
- 2 Push the display into the housing **from the back**, cable tab at 6 o'clock, until the glass stops against the front lip. It should go in by hand without forcing; the board sits in the wide pocket at the back, and the ribbon loop folds into the pocket below the tab.
- 3 Drop the back plate in, pegs out. It should rest on the board's brass studs and sit **flush with the base**. If it stands proud, the display isn't fully forward; find what's catching before going further.
- 4 Set the acrylic disc in the front pocket. It should sit flush with the rim.
- 5 Stand the unit on the jig with the pegs in the holes. The base should sit flat; hold it against the trunk lid's badge holder to check the fit and curve.

Adjusting the fit: every dimension is a value in hardware/source/haLo_housing.py. Change it and run python3 haLo_housing.py to write new STL and 3MF files.

Factory-gloss finish (optional): wet-sand 400 → 800 → 1500, adhesion promoter, filler primer, gloss black, 2K clear. Keep paint out of the lens pocket and off the base where the tape goes.

4 Firmware

No coding or software needed: it installs from a web page in about a minute. Do this on the bench, before the board goes into the housing.

Install it (once)

- 1 On a **computer** (Windows, Mac, Linux or Chromebook), open **Chrome** or **Edge**. Safari, Firefox and phones can't do this.
- 2 Go to **cakemasterjedi.github.io/haloweb**.
- 3 Plug the display into the computer with a **USB-C data cable**. Some cheap cables only charge; if nothing happens, try another cable.
- 4 Click **Install**. A box pops up with a list of ports: pick the one that appeared when you plugged in (it usually has *USB* or *CH343* in the name), click **Connect**, then **Install**.
- 5 Wait for *Installation complete* (about a minute). The display restarts with the emblem on it.

No port in the list? Install the **CH343** USB driver from wch-ic.com, or unplug, hold **BOOT**, plug back in, let go and try again.

Control it from your phone

- 1 Power the display. The first time, it makes its own Wi-Fi network (**EMBLEM-** and four characters) with a temporary password, and shows both on its screen.
- 2 Join it on your phone with that password. The control page opens by itself; if not, open the browser and go to **<http://192.168.4.1>**. On Android, tap **Stay connected** when it says the network has no internet.
- 3 It asks you to choose your own Wi-Fi password (and a name if you like), then restarts its Wi-Fi. Rejoin with your new password (write it down) and pick a design and brightness.

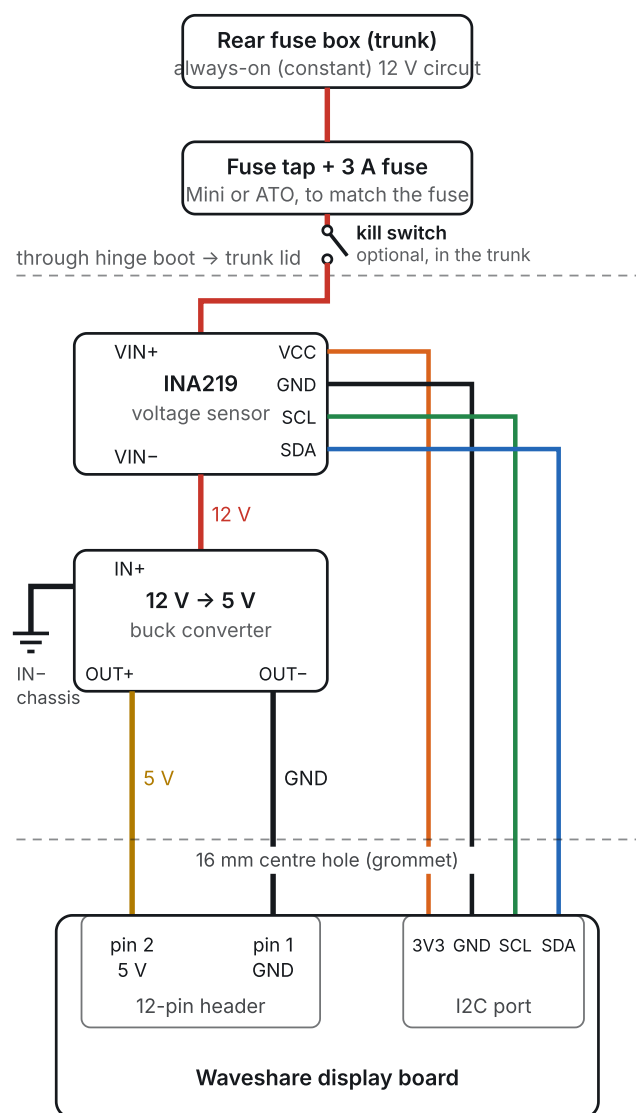
Set up the battery protection

On the **Power** tab, under **Battery protection**:

SETTING	WHAT TO CHOOSE
Voltage source	Car battery (INA219) . The live voltage appears once the INA219 is wired (section 5).
Battery protection	On , cutoff about 12.2 V . If the voltage stays below it for 30 seconds the display turns off. It checks every 10 minutes and comes back on at 0.3 V above the cutoff (engine running or charger on).
Calibrate	Type in what a multimeter reads across the battery (section 9).
Power mode → Turn off after	Leave it on Never unless you fitted a kill switch: once off, it stays off until the BOOT button on the board is pressed (sealed inside) or the kill switch cycles its power. Car show: Until the battery is low .

Updating later: download the update file from **cakemasterjedi.github.io/haloweb** on your phone, join the emblem's Wi-Fi, and install it under **Settings → Firmware**.

5 Wiring overview



12 V passes through the INA219 before the converter. All six display wires go through the 16 mm centre hole.

FROM	TO
● Fuse tap (3 A) → kill switch, if fitted	INA219 VIN+
● INA219 VIN-	Converter IN+ (red)
● Converter IN- (black)	Chassis ground stud
● Converter OUT+ (yellow)	12-pin header pin 2 (5 V)
● Converter OUT- (black)	12-pin header pin 1 (GND)
● INA219 VCC	I2C port 3V3
● INA219 GND	I2C port GND
● INA219 SCL	I2C port SCL (GPIO7)
● INA219 SDA	I2C port SDA (GPIO15)

Never

power the INA219 from the converter's 5 V (it would put 5 V on the ESP32's data pins), and never connect 12 V to the board's BAT connector.

Follow the pin labels printed on the board, not the wire colors of the supplied plug cables.

6 Build the display unit

- 1 **Make the cables.** Solder 28 AWG red and black to the 12-pin plug cable's pin 2 (5 V) and pin 1 (GND). Solder four 30 AWG wires to the 4-pin plug cable's 3V3, GND, SDA and SCL. Leave about 1 m on each, insulate every joint with heat shrink, and label the far ends.
- 2 **Plug in.** Plug the wires into the 12-pin header and the I2C port. Route them flat across the back of the board to the middle, staying clear of the brass studs, and twist them loosely into one bundle there.
- 3 **Conformal-coat the board.** With both plugs pushed fully home, mask the USB-C port and the BOOT and RESET buttons with tape and keep the coating off the front glass. Brush or spray a thin coat over the back of the board, the two connectors and the first centimetre of the wires. Let it dry as the can says and peel the masking.
- 4 **Optional heat spreader:** if you're fitting the aluminium disc and thermal pads, do it now, following Appendix A, and use the SPREADER back plate.
- 5 **Fit the display from the back.** Hold the housing face down on a soft cloth. Lower the display in, glass first and cable tab at 6 o'clock, and push it gently forward until the glass stops against the front lip all round. Fold the ribbon loop into the pocket below the tab. Don't force it: if it catches, find where and trim, rather than pressing on the glass.
- 6 **Fit the back plate.** Feed the wire bundle through the plate's centre hole from the board side, then drop the plate into the base, pegs out. It rests on the brass studs, flush with the base. Tape a mini silica gel pack to the back of the board first if there's room beside the studs.
- 7 **Seal the back.** Run a thin bead of black silicone around the plate's edge where it meets the housing, and fill the centre hole around the wires. Let it cure for 24 h.
- 8 **Final firmware check.** Power the board through the new wires (5 V supply or the converter) and confirm the screen, the voltage reading and the phone control page all work.
- 9 **Seal the glass edge.** From the front, run a thin bead of black silicone into the gap between the glass and the housing at the bottom of the lens pocket and let it cure.
- 10 **Bond the lens.** Clean the glass and the disc with isopropyl alcohol. Put a small pool of LOCA in the middle of the glass, lower the disc into the pocket from one edge so the glue spreads thin without bubbles, and press gently until it is flush with the rim. Wipe off the excess, then cure with UV light for the time given on the bottle.
- 11 **Seal the lens edge** with a fine bead of silicone in the 0.15 mm gap and let it cure.
- 12 **Water test before it goes on the car** (after 24 h of curing). Fit a test ring of tape on the base, stick the unit face up on a smooth tile and fill a bowl around it to 2 cm below the top of the rim for 30 minutes, then spray it with a garden hose for 2 minutes from all sides. Power it up and check: no fog under the lens, silica gel still dry. Peel the test tape and fit fresh tape for the car.

How it's sealed

Every way in is closed: the lens is bonded to the glass with LOCA and its edge is sealed with silicone; the glass edge is sealed from the front; the back plate's edge and the wires in its centre hole are silicone; on the car, a continuous tape ring seals the housing base and plate edge to the paint, and the 16 mm hole has

a grommet sealed with silicone; the board is conformal-coated with a silica gel pack inside for any trapped moisture.

Washing

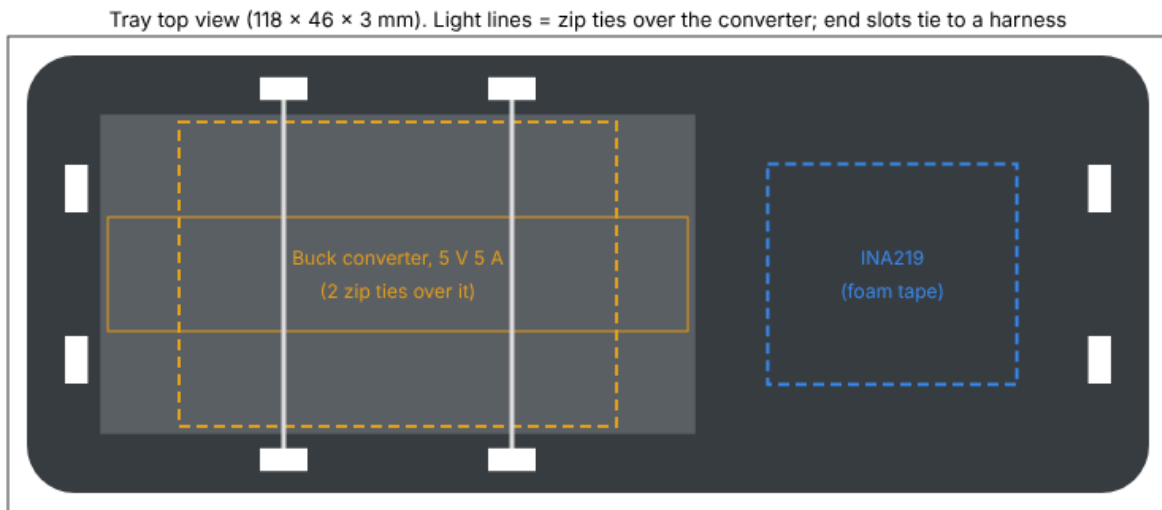
Hand washing, rain and touchless or brush car washes are fine. With a pressure washer, use a 25° or 40° (green or white) tip, stay at least 30 cm (1 ft) away and keep the jet moving; don't hold it on the rim's edge, and never use a 0° (red) tip. Wait 24 h after fitting before the first wash.

About ratings: this build follows the sealing practice used for IP67 (immersion) and pressure-washer-proof parts, and the water test above checks your unit. It isn't certified: an official IP67 or IPX9K (high-pressure, hot-water jet) rating needs testing by a lab.

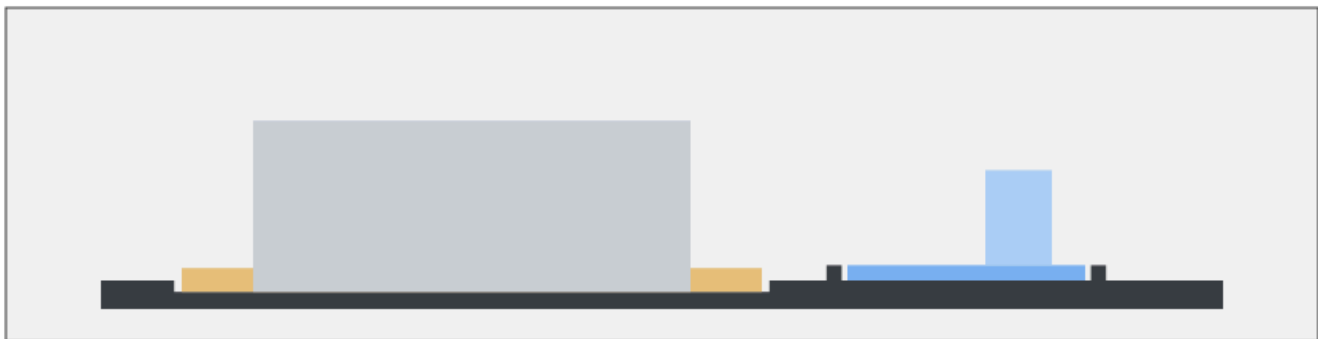
Don't use B-7000 on the lens

Its solvents can craze acrylic and fog the inside of the sealed housing.

7 Build the trunk-lid tray



Side view with parts in place



Converter strapped down with two zip ties; INA219 in its pocket on foam tape; end slots tie the tray to the trunk-lid harness.

- 1 Sit the converter's base and mounting ears in the pocket and strap it down with two zip ties through the slots beside it.
- 2 Solder wires to the INA219 with the pins or wires facing up, then stick it in its pocket with foam tape.
- 3 Wire the 12 V path: incoming 12 V to VIN+, VIN- to the converter's red input lead, converter's black input lead to a ring terminal for ground. Use solder-seal connectors and heat shrink everywhere.
- 4 Join the converter's yellow and black output leads to the display's power pair, and the INA219's VCC, GND, SDA and SCL to the display's sensor wires. Each wire goes to its match: the INA219's GND to the sensor GND wire.

8 Install on the car

Remove the badge and prepare the paint

- 1 Open the trunk and remove the trunk lid's inner trim panel (it clips off) to reach the back of the badge and the lid harness.
- 2 Release the badge's pegs from inside if you can reach them, or pry the badge off from outside with a plastic tool. Leave the two grommets in the holes; the plate's pegs locate in them.
- 3 Check the grommets sit flush with the paint. If their heads stand proud, the plate will rock; trim them flush.
- 4 **Drill the wire hole.** Check behind the panel first for braces, wiring, the camera and the latch. Mark the exact centre between the two peg holes (28 mm from each), centre-punch it through a piece of painter's tape and drill 16 mm (5/8") with a step drill. Deburr, vacuum the chips, paint the bare edge with touch-up paint and let it dry, then fit the 5/8" grommet.
- 5 Clean the area with isopropyl alcohol, then wipe adhesion promoter where the tape ring will sit.

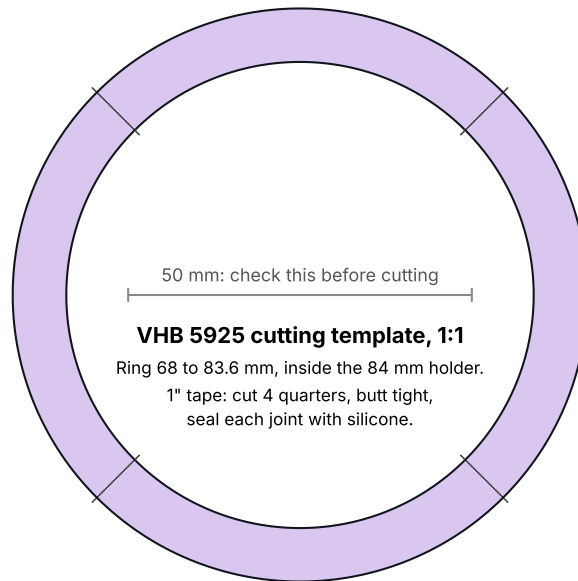
Run the 12 V feed

- 1 Find an **always-on** fuse in the trunk fuse box (right side, behind the trim cover). Lock the car and wait about 20 minutes for it to go to sleep. Clip a test light or the multimeter's black lead to a bare metal bolt, then touch the two little metal test points on top of each fuse. A fuse that still lights up (or reads about 12 V) on one side is always on.
- 2 Good ones to test first are the **12 V socket** fuses (shown as "cigarette lighter / 12 V socket" on the fuse chart) and slots for features your car doesn't have. Never tap anything labelled DME, IVM, airbag, seatbelt, fuel pump, steering column or DSC.
- 3 Pull the fuse you chose and check its size: **Mini** (small) or **ATO** (standard, larger). Use the matching fuse tap and a 3 A fuse of the same size. Disconnect the battery. Fit the fuse tap with the original fuse in the lower slot and a 3 A fuse in the upper slot, turned so the new lead is on the load side.
- 4 **Optional kill switch:** splice it into the red lead right after the fuse tap and leave it where you can reach it, behind the trunk side trim flap near the fuse box. An inline rocker switch on a pigtail needs no mounting.
- 5 Ring-terminal the ground to a factory ground stud on bare metal.
- 6 Run the red 20 AWG lead along the factory harness to the trunk hinge and through the rubber boot into the lid, inside split loom, wrapped in Tesa tape. Keep it clear of the hinge, latch and any sharp edges.

Mount the tray and the display

- 1 Zip-tie the tray to the harness inside the lid (or VHB it to a clean inner panel), somewhere you can reach later. Connect the 12 V feed to the INA219's VIN+.
- 2 **Cut the tape.** Print the template below at 100 % scale (check the 50 mm line), lay the 0.64 mm VHB over it and cut one continuous ring. With 1" tape, cut the four quarters, butt them tight and seal each joint with a dab of silicone.

- 3 Stick the ring on the base so it covers the housing base and the edge of the back plate all round.
- 4 Feed the wires through the grommet into the lid and seal the grommet around them with silicone. Peel the liner, line the pegs up with the factory holes (cable tab at 6 o'clock), set the housing into the badge holder and press it firmly all round for 30 seconds.
- 5 From inside, join the wires to the tray as in step 7.4.
- 6 Secure the loose wiring inside the lid with tape and zip ties, then refit the trim panel. Leave the car for 24 h before washing so the tape reaches full strength.



VHB cutting template, 1:1. Print at 100 % (no "fit to page") and check that the line measures 50 mm. Also in the hardware folder as VHB_tape_template.svg.

9 Test and calibrate

- 1 Reconnect the battery and wake the car. The display should boot within a few seconds.
- 2 With the engine off, measure the battery with a multimeter. Join the emblem's Wi-Fi on your phone, open the **Power** tab, type the reading into **Calibrate** and tap Calibrate. The page and the screen now match the meter.
- 3 Start the engine: the reading should rise to about 13.6–14.6 V and show "Charging".
- 4 Lock the car and leave it. The display should stay on. If you want to test the battery protection, raise the cutoff above the battery voltage for a moment: after 30 seconds the display turns off, and it comes back within 10 minutes of starting the engine. Set the cutoff back afterwards.
- 5 If you fitted the kill switch, check it turns the display fully off.

READING	WHAT IT MEANS
13.2 V and up	Engine running, alternator charging
12.4–12.8 V	Healthy battery at rest
12.0–12.4 V	Partly discharged
Under 12.0 V	Low; under 11.8 V for a minute triggers the optional backlight-off action

10 Troubleshooting

PROBLEM	CHECK
Display doesn't turn on	Fuse tap direction and 3 A fuse; 5 V at the converter output; pin 1 / pin 2 on the 12-pin header the right way round.
Voltage shows "No sensor"	INA219 VCC on 3.3 V (not 5 V); SDA/SCL not swapped; plugged into the I2C port, not the UART port.
Fog inside the lens	A seal is leaking or the housing was closed on a humid day. Replace the silica gel and reseal the glass edge. Assemble on a dry day.
Screen goes dark on a hot day, comes back later	That's the heat guard: it dims the screen from about 70 °C on the chip and turns it off above 85 °C until it cools to 75 °C. The Power tab shows the chip temperature and the hottest it has been. The ESP32-S3R8 on this board is rated to 65 °C air temperature and the LCD to 70 °C (80 °C when off).
Display goes dark when the car is off	The tapped fuse is switched, not always-on: pick another (section 8). Or the battery is under the cutoff; charge it. Or Normal mode auto-off is on: turn it off.
Battery runs down while parked	Raise the battery protection cutoff, or use the kill switch when the car will sit. Asleep, the display still draws a little.

Can't find the emblem's Wi-Fi	The display may be off (battery protection or auto-off). Turn it off and on with the kill switch or fuse. Look for the name you chose at setup (or EMBLEM-xxxx on a new display).
Web installer finds no port	Use Chrome or Edge on a computer and a USB-C data cable. Hold BOOT while plugging in, then try again.
Display won't go fully in, or the back plate stands proud	Something on the board or the cable tab is catching. Find the spot, trim it, or raise <code>disp_clear</code> , <code>chin_hw</code> or <code>board_grow</code> in <code>halo_housing.py</code> and reprint. The plate should sit flush with the base when the glass touches the front lip.
Lens sits low or proud	Lenses vary in thickness. Set <code>lens_t</code> in <code>halo_housing.py</code> to your disc's measured thickness and reprint.

A Heat spreader and thermal pads (optional)

In full sun a black trunk lid gets very hot, and the board adds about 1 W of its own. This option moves that heat into the trunk lid instead of letting it build up inside the housing. It keeps the same height and the same tape-ring seal, because everything sits inside the ring.

- **Print** `HALO_88_BACK_PLATE_SPREADER.3mf` instead of the plain plate. It has a 46 mm window and a 50.6 mm, 0.4 mm-deep recess on the paint side.
- **Cut the disc:** print `HALO_Spreader_Disc_Template_1to1.pdf` at 100 %, tape it to 0.016" (0.4 mm) aluminium sheet, cut the 50 mm disc with tin snips and step-drill the 16 mm centre hole so the wires still pass. File the edges smooth.
- **Insulate it:** cover the disc's board side with Kapton tape, leaving bare spots only where pads sit, so nothing on the board can short to the metal.
- **Pads:** with the display and plate dry-fitted, measure from the disc to the hot parts on the back of the board (ESP32 module, regulators, backlight driver). Use soft, electrically insulating gap pads (3–6 W/mK) about 30 % thicker than the gap so they compress lightly. Stick them on the hot parts.
- **Fit:** glue the disc into the recess with a few dots of silicone at its edge, flush with the plate, then fit the plate as in step 6.6. The pads press lightly on the disc through the window.
- **On the car:** the badge seat dips about 0.86 mm at the centre, so there's a 1–1.5 mm gap under the disc. Fill it with a 1.5 mm soft thermal pad ring (50 mm across, 16 mm hole) on the disc's paint side so the disc touches the paint through the pad.

What to expect: my estimate is a few degrees to about 10 °C cooler in full sun; it can't make the display cooler than the lid itself. The Power tab's hottest-temperature record shows the real difference.

Other ways to keep it cooler

- Turn brightness down in summer; the backlight is most of the heat.
- Use darker designs during the day; a mostly black screen runs cooler than a bright one.
- The heat guard (section 10) protects it on the hottest days by dimming from about 70 °C on the chip and turning the screen off above 85 °C.